

Biostat 203B Homework 1

Due Jan 24, 2025 @ 11:59PM

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Display machine information for reproducibility:

```
sessionInfo()
```

```
R version 4.4.2 (2024-10-31 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26100)
```

```
Matrix products: default
```

```
locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8
```

```
time zone: America/Los_Angeles
tzcode source: internal
```

attached base packages:

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

loaded via a namespace (and not attached):

```
[1] compiler_4.4.2    fastmap_1.2.0     cli_3.6.3         tools_4.4.2
[5] htmltools_0.5.8.1 rstudioapi_0.16.0 yaml_2.3.10       tinytex_0.53
[9] rmarkdown_2.28    knitr_1.48        jsonlite_1.8.9    xfun_0.50
[13] digest_0.6.37     rlang_1.1.4       evaluate_1.0.3
```

0.1 Q1. Git/GitHub

GitHub created. Repository Created.

0.2 Q2. Data ethics training

Display the verification links to your completion report and completion certificate here.

Completion Report

Here is the Completion report in full: citiprogram.org/verify/?kca45a062-ea0a-4df1-9e13-71ca2b20e7f0-67410439

Completion Certificate

Here is the Certificate link in full: citiprogram.org/verify/?wad4b3eab-a4ed-4be4-b940-dce1260f4f13-67410439

0.3 Q3. Linux Shell Commands

1.

```
# Create mirror location for mimic dataset
ln -s "C:/Users/boyle/Downloads/mimic-iv-3.1/mimic-iv-3.1" ~/mimic
```

```
# content of mimic folder
ls -l ~/mimic
```

2. Display the contents in the folders hosp and icu using Bash command `ls -l`.

```
# content of hosp folder in mimic folder
ls -l ~/mimic/hosp
```

Contains hospital level data for patients: labs, micro, and electronic medication administration.

```
# content of icu folder in mimic folder
ls -l ~/mimic/icu
```

Contains ICU level data. These are the event tables, and are identical in structure to MIMIC-III (chartevents, etc)

Why are these data files distributed as `.csv.gz` files instead of `.csv` (comma separated values) files?

Typically this is done to reduce file size, as this dataset is rather large.

3. Briefly describe what Bash commands `zcat`, `zless`, `zmore`, and `zgrep` do.

`zcat`: Allows user to view a compressed file without uncompressing the file

`zless`: Can view compressed files interactively, going page by page.

`zmore`: same as `zless` but has reduced navigation options.

`zgrep`: can find patterns in compressed files.

4. (Looping in Bash) What's the output of the following bash script?

```
for datafile in ~/mimic/hosp/{a,l,pa}*.gz
do
    ls -l $datafile
done
```

This prints out the file location of gz files that start with the letters “a”, “l”, and “pa.”

Display the number of lines in each data file using a similar loop. (Hint: combine linux commands `zcat` < and `wc -l`.)

```
for datafile in ~/mimic/hosp/{a,l,pa}*.gz
do
    zcat $datafile | wc -l
done
```

The result printed was: 546029

5. Display the first few lines of `admissions.csv.gz`.

```
zcat ~/mimic/hosp/admissions.csv.gz | head
```

Here is what was printed: subject_id,hadm_id,admittime,dischtime,deathtime,admission_type,admit_provider
10000032,22595853,2180-05-06 22:23:00,2180-05-07 17:15:00,,URGENT,P49AFC,TRANSFER
FROM HOSPITAL,HOME,Medicaid,English,WIDOWED,WHITE,2180-05-06 19:17:00,2180-
05-06 23:30:00,0 10000032,22841357,2180-06-26 18:27:00,2180-06-27 18:49:00,,EW EMER.,P784FA,EMERGENCY
ROOM,HOME,Medicaid,English,WIDOWED,WHITE,2180-06-26 15:54:00,2180-06-26
21:31:00,0 10000032,25742920,2180-08-05 23:44:00,2180-08-07 17:50:00,,EW EMER.,P19UTS,EMERGENCY
ROOM,HOSPICE,Medicaid,English,WIDOWED,WHITE,2180-08-05 20:58:00,2180-08-06
01:44:00,0 10000032,29079034,2180-07-23 12:35:00,2180-07-25 17:55:00,,EW EMER.,P06OTX,EMERGENCY
ROOM,HOME,Medicaid,English,WIDOWED,WHITE,2180-07-23 05:54:00,2180-07-23
14:00:00,0 10000068,25022803,2160-03-03 23:16:00,2160-03-04 06:26:00,,EU OBSERVA-
TION,P39NWO,EMERGENCY ROOM,,English,SINGLE,WHITE,2160-03-03 21:55:00,2160-
03-04 06:26:00,0 10000084,23052089,2160-11-21 01:56:00,2160-11-25 14:52:00,,EW EMER.,P42H7G,WALK-
IN/SELF REFERRAL,HOME HEALTH CARE,Medicare,English,MARRIED,WHITE,2160-
11-20 20:36:00,2160-11-21 03:20:00,0 10000084,29888819,2160-12-28 05:11:00,2160-12-28
16:07:00,,EU OBSERVATION,P35NE4,PHYSICIAN REFERRAL,,Medicare,English,MARRIED,WHITE,2160-
12-27 18:32:00,2160-12-28 16:07:00,0 10000108,27250926,2163-09-27 23:17:00,2163-09-28
09:04:00,,EU OBSERVATION,P40JML,EMERGENCY ROOM,,English,SINGLE,WHITE,2163-
09-27 16:18:00,2163-09-28 09:04:00,0 10000117,22927623,2181-11-15 02:05:00,2181-11-15
14:52:00,,EU OBSERVATION,P47EY8,EMERGENCY ROOM,,Medicaid,English,DIVORCED,WHITE,2181-
11-14 21:51:00,2181-11-15 09:57:00,0

How many rows are in this data file, excluding the header line?

```
zcat ~/mimic/hosp/admissions.csv.gz | tail -n +2 | wc -l
```

This printed the result: 546028

Each `hadm_id` identifies a hospitalization. How many hospitalizations are in this data file?

```
zcat ~/mimic/hosp/admissions.csv.gz | cut -d',' -f2 | tail -n +2 | sort | uniq | wc -l
```

This printed: 546028 unique hospitalization ids

How many unique patients (identified by `subject_id`) are in this data file?

```
zcat ~/mimic/hosp/admissions.csv.gz | cut -d',' -f1 | tail -n +2 | sort | uniq | wc -l
```

This printed: 223452 unique subject ids.

Do they match the number of patients listed in the `patients.csv.gz` file?

```
zcat ~/mimic/hosp/patients.csv.gz | cut -d',' -f1 | tail -n +2 | sort | uniq | wc -l
```

No they do not. This bash command printed: 364627 unique subject ids.

6. What are the possible values taken by each of the variable `admission_type`, `admission_location`, `insurance`, and `ethnicity`?

Admission Type

```
zcat ~/mimic/hosp/admissions.csv.gz | cut -d',' -f6 | tail -n +2 | sort | uniq
```

This printed the following variables: AMBULATORY OBSERVATION DIRECT EMER. DIRECT OBSERVATION ELECTIVE EU OBSERVATION EW EMER. OBSERVATION ADMIT SURGICAL SAME DAY ADMISSION URGENT

Admission Location

```
zcat ~/mimic/hosp/admissions.csv.gz | cut -d',' -f8 | tail -n +2 | sort | uniq
```

This printed the following locations: AMBULATORY SURGERY TRANSFER CLINIC REFERRAL EMERGENCY ROOM INFORMATION NOT AVAILABLE INTERNAL TRANSFER TO OR FROM PSYCH PACU PHYSICIAN REFERRAL PROCEDURE SITE TRANSFER FROM HOSPITAL TRANSFER FROM SKILLED NURSING FACILITY WALK-IN/SELF REFERRAL

Race

```
zcat ~/mimic/hosp/admissions.csv.gz | cut -d',' -f13 | tail -n +2 | sort | uniq
```

This printed the following ethnicities: AMERICAN INDIAN/ALASKA NATIVE ASIAN ASIAN - ASIAN INDIAN ASIAN - CHINESE ASIAN - KOREAN ASIAN - SOUTH EAST ASIAN BLACK/AFRICAN BLACK/AFRICAN AMERICAN BLACK/CAPE VERDEAN BLACK/CARIBBEAN ISLAND HISPANIC OR LATINO HISPANIC/LATINO - CENTRAL AMERICAN HISPANIC/LATINO - COLUMBIAN HISPANIC/LATINO - CUBAN HISPANIC/LATINO - DOMINICAN HISPANIC/LATINO - GUATEMALAN HISPANIC/LATINO - HONDURAN HISPANIC/LATINO - MEXICAN HISPANIC/LATINO - PUERTO RICAN HISPANIC/LATINO - SALVADORAN MULTIPLE RACE/ETHNICITY NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER OTHER PATIENT DECLINED TO ANSWER PORTUGUESE SOUTH AMERICAN UNABLE TO OBTAIN UNKNOWN WHITE WHITE - BRAZILIAN WHITE - EASTERN EUROPEAN WHITE - OTHER EUROPEAN WHITE - RUSSIAN

Also report the count for each unique value of these variables in decreasing order.

Admission Type

```
zcat ~/mimic/hosp/admissions.csv.gz | cut -d',' -f6 | tail -n +2 | sort | uniq -c | sort -nr
```

This prints: 177459 EW EMER. 119456 EU OBSERVATION 84437 OBSERVATION ADMIT
54929 URGENT 42898 SURGICAL SAME DAY ADMISSION 24551 DIRECT OBSERVA-
TION 21973 DIRECT EMER. 13130 ELECTIVE 7195 AMBULATORY OBSERVATION

Admission Location

```
zcat ~/mimic/hosp/admissions.csv.gz | cut -d',' -f8 | tail -n +2 | sort | uniq -c | sort -nr
```

This prints: 244179 EMERGENCY ROOM 163228 PHYSICIAN REFERRAL 56227 TRANS-
FER FROM HOSPITAL 42365 WALK-IN/SELF REFERRAL 12965 CLINIC REFERRAL
8518 PROCEDURE SITE 6317 TRANSFER FROM SKILLED NURSING FACILITY 5837
INTERNAL TRANSFER TO OR FROM PSYCH 5734 PACU 402 INFORMATION NOT
AVAILABLE 255 AMBULATORY SURGERY TRANSFER 1

Race

```
zcat ~/mimic/hosp/admissions.csv.gz | cut -d',' -f13 | tail -n +2 | sort | uniq -c | sort -nr
```

This prints: 336538 WHITE 75482 BLACK/AFRICAN AMERICAN 19788 OTHER 13972
WHITE - OTHER EUROPEAN 13870 UNKNOWN 10903 HISPANIC/LATINO - PUERTO
RICAN 8287 HISPANIC OR LATINO 7809 ASIAN 7644 ASIAN - CHINESE 6597 WHITE -
RUSSIAN 6205 BLACK/CAPE VERDEAN 6070 HISPANIC/LATINO - DOMINICAN 3875
BLACK/CARIBBEAN ISLAND 3495 BLACK/AFRICAN 3478 UNABLE TO OBTAIN 2162
PATIENT DECLINED TO ANSWER 2082 PORTUGUESE 1973 ASIAN - SOUTH EAST
ASIAN 1886 WHITE - EASTERN EUROPEAN 1858 HISPANIC/LATINO - GUATEMALAN
1661 ASIAN - ASIAN INDIAN 1526 WHITE - BRAZILIAN 1320 HISPANIC/LATINO -
SALVADORAN 1247 AMERICAN INDIAN/ALASKA NATIVE 920 HISPANIC/LATINO
- COLUMBIAN 883 HISPANIC/LATINO - MEXICAN 774 SOUTH AMERICAN 725 HIS-
PANIC/LATINO - HONDURAN 664 ASIAN - KOREAN 641 HISPANIC/LATINO - CUBAN
603 HISPANIC/LATINO - CENTRAL AMERICAN 596 MULTIPLE RACE/ETHNICITY
494 NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER

7. The `icusays.csv.gz` file contains all the ICU stays during the study period. How many ICU stays, identified by `stay_id`, are in this data file?

```
zcat ~/mimic/icu/icustays.csv.gz | cut -d',' -f3 | tail -n +2 | sort | uniq | wc -l
```

This prints: 94458 entries for stay_id

How many unique patients, identified by subject_id, are in this data file?

```
zcat ~/mimic/icu/icustays.csv.gz | cut -d',' -f1 | tail -n +2 | sort | uniq | wc -l
```

We find 65366 entries in subject_id.

8. *To compress, or not to compress. That's the question.* Let's focus on the big data file `labevents.csv.gz`. Compare compressed gz file size to the uncompressed file size.

```
gzip -l ~/mimic/hosp/labevents.csv.gz
```

This prints the size for compressed and uncompressed version of labevents.csv.gz

We find the compressed file is 2592909134 bytes and the uncompressed file size is 18402851720 bytes.

Compare the run times of `zcat < ~/mimic/labevents.csv.gz | wc -l` versus `wc -l labevents.csv`.

```
time zcat < ~/mimic/hosp/labevents.csv.gz | wc -l
```

My system had the following results: real 1m6.293s user 1m4.812s sys 0m9.109s

```
time wc -l ~/mimic/hosp/labevents.csv
```

After unpacking the compressed files, my system had the following time: real 0m0.049s user 0m0.015s sys 0m0.015s

Discuss the trade off between storage and speed for big data files.

The increased speed with uncompressed large data files can be 100s of factors faster to work with. However these data files can be incredibly large. It may become difficult to work with multiple large datasets at once on a single system.

0.4 Q4. Who's popular in Price and Prejudice

- 1.

```
wget -nc http://www.gutenberg.org/cache/epub/42671/pg42671.txt
```

Explain what `wget -nc` does.

Wget is a command in linux that downloads files from url codes. The `-nc` option stands for “no clobber” which is used to prevent overwriting files that already exist.

Complete the following loop to tabulate the number of times each of the four characters is mentioned using Linux commands.

```
wget -nc http://www.gutenberg.org/cache/epub/42671/pg42671.txt
for char in "Elizabeth" "Jane" "Lydia" "Darcy"
do
    echo "$char:"
    grep -o -i "$char" pg42671.txt | wc -l
done
```

This prints:

Elizabeth: 634 Jane: 293 Lydia: 171 Darcy: 418

2. What’s the difference between the following two commands?

```
echo 'hello, world' > test1.txt
```

and

```
echo 'hello, world' >> test2.txt
```

The first line of code ‘>’ overwrites the contents of test1.txt. If there was any data in test1.txt it would of been completely replaced with the string ‘hello, world’

The 2nd line of code applies an appending operator ‘>>’ to the .txt file. This means if test2.txt already had data in it, the string ‘hello, world’ would have been added to the end of the file.

3. Using your favorite text editor (e.g., `vi`), type the following and save the file as `middle.sh`:

```
#!/bin/sh
# Select lines from the middle of a file.
# Usage: bash middle.sh filename end_line num_lines
head -n "$2" "$1" | tail -n "$3"
```


Using `chmod` to make the file executable by the owner, and run

```
./middle.sh pg42671.txt 20 5
```

Explain the output. Explain the meaning of "\$1", "\$2", and "\$3" in this shell script. Why do we need the first line of the shell script?

This output takes the first 20 lines of the file `pg42671.txt` and then prints the last 5 lines from the first 20. The "\$1", "\$2", and "\$3" are known as positional parameters. They represent the values we provide after the script is run. In this case \$1 represents the filename (`pg42671.txt`), \$2 represents the number of "head" lines we want to look at, and \$3 represents the number of "tail" lines we want to display.

The first line of the shell script tells the operating system what interpreter we need to use to run the script "middle.sh." If we did not do this, we could use the wrong interpreter for the script which will cause the executable to run incorrectly and may lead to many errors.

0.5 Q5. More fun with Linux

Try following commands in Bash and interpret the results: `cal`, `cal 2025`, `cal 9 1752` (anything unusual?), `date`, `hostname`, `arch`, `uname -a`, `uptime`, `who am i`, `who`, `w`, `id`, `last | head`, `echo {con,pre}{sent,fer}{s,ed}`, `time sleep 5`, `history | tail`.

`Cal`: prints out the current calendar month and year

`Cal 2025`: prints out all calendar months of the specified year

`Cal 9 1752`: should print out the month of september (9) in the year 1752 For some reason "cal 9 1752" prints a calendar of September with missing days. This is because this is when people switched from the Julian calendar to the Gregorian calendar.

`Date`: prints out the exact date and time that the OS has.

`Hostname`: prints the host name of the linux user. In my case this is the name of the PC.

`Arch`: prints out the architecture of the cpu.

`uname -a`: prints out all (-a for all) information about the system linux is being run on.

`Uptime`: prints how long the system has been running since it was last booted. Also shows number of users and the load average, which is a measurement of the average number of processes waiting to be executed by the cpu.

`who am i`: prints the username of the current logged in user

`Who`: prints the information about all users who are logged into the system. Displays the username, what terminal and login time.

W: an expanded command on “who” this prints out a wide variety of information about users on the system. This includes current activity and the time they’ve been idle as well as the system load.

Id: prints the users uid (user id) and the username associated with that id, the GID which is the user’s primary group (usually this is the same as the uid) and other administrative or otherwise ids associated with the user.

Last | head : provides the last logged in users on the system. Prints a history of user logins and shows when users logged in and out. Also shows the duration of the user’s sessions and the IP address. Head will provide the first 10 lines of the “last” output

echo {con,pre}{sent,fer}{s,ed}: this line takes advantage of the brace expansion in linux. Linux will generate multiple strings passed on a pattern specified by us.

time sleep 5: the time command measures the amount of time another command takes. The sleep command tells the OS to pause or delay an execution for a specified amount of time, in this case 5 seconds.

history | tail: prints the history of commands ran in the current instance of this shell. Tail specifies to only print the last 10 arguments of history.

0.6 Q6. Book

1. Git clone the repository <https://github.com/christophergandrud/Rep-Res-Book> for the book *Reproducible Research with R and RStudio* to your local machine. Do **not** put this repository within your homework repository `biostat-203b-2025-winter`.
2. Open the project by clicking `rep-res-3rd-edition.Rproj` and compile the book by clicking **Build Book** in the **Build** panel of RStudio. (Hint: I was able to build `git_book` and `epub_book` directly. For `pdf_book`, I needed to add a line `\usepackage{hyperref}` to the file `Rep-Res-Book/rep-res-3rd-edition/latex/preabmle.tex`.)

The point of this exercise is (1) to obtain the book for free and (2) to see an example how a complicated project such as a book can be organized in a reproducible way. Use `sudo apt install PKGNAME` to install required Ubuntu packages and `tlmgr install PKGNAME` to install missing TeXLive packages.

For grading purpose, include a screenshot of Section 4.1.5 of the book here.

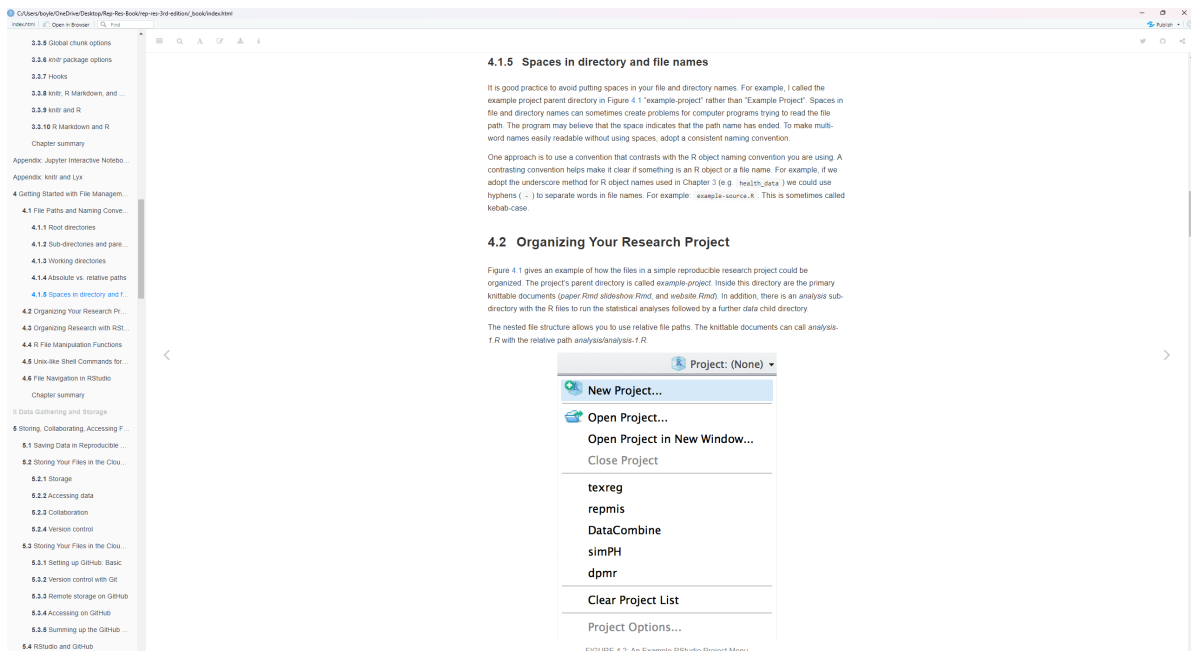


Figure 1: My Image